Evolving Reputation for Commitment: Understanding Inflation and Inflation Expectations (Referee Report)

Summary

This paper contributes to the literature on monetary policy credibility and inflation expectations. On the theoretical side, the authors develop a novel framework for analyzing optimal monetary policy when private agents learn about policymaker type (committed versus opportunistic) and both types of policymakers optimize. The central innovation is that the committed policymaker strategically manages private-sector learning by choosing policies that accelerate reputation-building when credibility is low. Using insights from mechanism design, the authors recast the equilibrium of a dynamic game with incomplete information as a principal-agent problem and provide a recursive solution with only three state variables: reputation, a cost-push shock, and a pseudo-state capturing past commitments. This represents a technical advance over prior work that either assumed one policymaker type follows a fixed rule or abstracted from the strategic dimension of reputation management.

On the empirical side, the authors develop a state extraction methodology using the term structure of Survey of Professional Forecasters inflation expectations and an unscented Kalman filter. They exploit the insight that short-term forecasts respond more to transitory cost-push shocks while longer-term forecasts reflect the persistent reputation state. The estimated reputation path exhibits a striking pattern, declining throughout the 1970s to near zero by 1980, then gradually recovering after 1981. This pattern aligns with conventional narratives about the Great Inflation and the Volcker disinflation. Notably, the model-implied inflation series tracks actual U.S. inflation despite inflation not being targeted in estimation. The authors also provide reduced-form evidence that the sensitivity of long-term forecast revisions to inflation surprises varies over time in a manner consistent with their model's predictions about strategic reputation management, lending empirical support to the mechanism that distinguishes their framework from simpler learning models.

1 Major Comments

- **Alternative Explanations for Observed Patterns:** The authors argue that their framework differs from standard regime-switching models by deriving policy endogenously from optimization rather than specifying exogenous policy rules. This is a genuine theoretical contribution that is one of the major strengths of the paper in my opinion. In their model, the committed policymaker strategically chooses policy to accelerate private-sector learning, generating a policy gap that varies with reputation: larger when reputation is low, smaller when reputation is high. This feedback from beliefs to policy, and the resulting nonlinear learning dynamics, would not arise in models with fixed policy rule differences across regimes. The forecast revision regressions in Section 6.3, which show that $\rho(1-\rho)^2$ weighting outperforms simpler specifications, represent a useful step toward testing this distinctive prediction.

However, this theoretical distinction does not fully resolve the empirical identification challenge. The estimated reputation path is extracted using the maintained model, so the close fit between model implications and data does not discriminate between the commitment/reputation interpretation and alternatives such as learning about the inflation target, policy preferences, or Phillips curve parameters. These alternative learning models could potentially generate similar dynamics in inflation expectations without invoking the commitment mechanism. To strengthen the identification argument, the authors could consider several approaches:

1. Derive additional predictions that uniquely distinguish the model from target-learning or preference-learning alternatives.
2. Estimate a competing specification using the same data and compare model fit, e.g. models of the type surveyed in Eusepi and Preston (2018).
3. Use non-expectations data such as forecast dispersion, inflation risk premia, or narrative measures of Fed credibility, to independently corroborate the estimated reputation path. See e.g. Reis (2022) for some concrete ideas about how to do this for measures of dispersion.
4. Examine whether actual Fed policy was systematically more aggressive when estimated reputation was low, as the strategic reputation-building mechanism predicts.

Any or all of these approaches would help establish that the specific mechanism of learning about commitment type, accelerated by strategic policy choices, is operative in the data rather than being one of several observationally similar interpretations.

- **Calibration and Estimation of the Model:** Why are hidden states estimated using only one- and three-quarter-ahead SPF forecasts of GDP deflator inflation? I understand the desire to leave untargeted series to help validate the model fit, but especially in models involving learning and belief formation, each additional series provides valuable information about the different mechanisms at play. What happens if all of the short-run, medium-run, and long-run forecasts of inflation, along with realized inflation, are used in the estimation? If long-run expectations are not used directly in estimating the hidden states, they could still serve as untargeted series to validate the model. How well does the baseline model capture the evolution of long-run inflation expectations over time?

The SPF contains not only one- to four-quarter-ahead forecasts of GDP deflator inflation, but also one-year-forward forecasts (constructed using the codes PGDPA and PGDPB in the survey). Additionally, for CPI inflation starting in 1981:Q3, one- to four-quarter-ahead and one- and two-year-forward forecasts are available. Ten-year average forecasts of CPI inflation are available starting in 1991:Q4, and five-year average and three-year-forward forecasts are available starting in 2005:Q3. If forecasts of GDP deflator inflation and CPI inflation are to be treated equivalently (see my next set of comments on long-run SPF inflation forecasts), then all of these series could be brought to bear on the estimation. Data on the output gap, whether as estimated by the CBO or filtered from real GDP data using, e.g., Hamilton (2018), would also help inform the estimates of the hidden states.

Building on these points, why not jointly estimate some of the model parameters along with the hidden states? Some of the assumed parameter values are quite extreme. The authors adopt an annualized inflation target of 6%, but empirical estimates of the long-run inflation trend over the sample period vary dramatically. A target of 6% seems far too high for the last three decades of inflation data. See, e.g., Del Negro et al. (2017). The Fed adopted an explicit PCE inflation target of 2% in 2012, and professional forecasters' long-run expectations of inflation have hovered between roughly 2–3% for nearly three decades.

The slope of the Phillips curve determines the output cost of disinflation and thus the severity of the time-inconsistency problem. The chosen value of $\kappa = 0.08$, corresponding to 0.02 when expressed in annualized terms, likely has a significant impact on the quantitative findings. Estimates from the literature vary widely; see, e.g., Hazell et al. (2022), who report a slope of 0.0062 for annual data, almost four times smaller than the value adopted here. Similar concerns apply to the output target and, especially, to the parameters governing reputation inheritance and cost-push shock dynamics. At a minimum, the authors should examine the sensitivity of their inferred reputation and cost-push shock measures to alternative configurations of structural parameters.

Lastly, the authors use a version of the unscented Kalman filter to perform the filtering of hidden states. They claim that this approach "has been shown to work well in nonlinear regime switching models" (p. 21), but it is not obvious ex ante that it will perform well for the specific model in this paper. I would appreciate a comparison of the hidden states estimated from a particle filter (with a large number of particles) against those estimated using the unscented Kalman filter. Given that the authors fix the model parameters, this exercise should be relatively straightforward and would help alleviate concerns that the results are partly driven by the choice of filtering method. This comparison could be relegated to an appendix.

- **Treatment of Long-Run SPF Inflation Forecasts:** In Section 6.3, the authors consider a validation exercise that uses long-term inflation forecasts from the SPF. I have several concerns about the treatment of the data in this section:

  1. The primary data used in estimating the model are short-run SPF forecasts of GDP deflator inflation; however, the long-run forecasts used here are of CPI inflation. I understand that only long-run forecasts of CPI inflation are available, but CPI inflation and GDP deflator inflation have somewhat different time-series properties and exhibit a correlation of only 0.79 to 0.87 (depending on whether end-of-quarter or average inflation is used) in the post-World War II period. It is not clear whether these forecasts are directly comparable. Furthermore, the object that the Fed has been explicitly targeting since 2012 is PCE inflation. One possible solution would be to use the period of overlapping short-run SPF forecasts of GDP deflator inflation and CPI inflation to infer implied values of long-run GDP deflator forecasts via a state-space measurement model.

  2. Ten-year-ahead forecasts of inflation are not the same object captured by the model's equation (19) on p. 27. The long-run expectations measured in the SPF reflect average inflation over the subsequent ten years, which includes short-run inflation, rather than a forecast of quarter-over-quarter inflation ten years from now. In periods when inflation differs significantly from the perceived inflation target, such as during COVID-19, this can lead to substantial deviations between the two concepts.

  3. The ten-year-ahead forecasts do not capture the same object over the course of a calendar year because they are fixed-event rather than fixed-horizon forecasts. They include inflation in the calendar year in which the forecast is made, so forecasts made later in the year contain a larger fraction of realized inflation. Note that this also implies forecast revisions constructed from these objects, which are used in regression-based tests of the model, are not consistently defined. This issue needs to be properly accounted for. See, e.g., Appendix A.1.3 in Aruoba (2020) for a discussion of this point and how to adjust the forecasts accordingly.

- **Discussion of the COVID-19 Period:** The estimated reputation remains high (above 0.8) throughout the recent inflation surge (2021–2024), yet Figure 3 shows that the cost-push shock rises sharply. The paper would benefit from an explicit discussion of how the model interprets this episode. Did the post-COVID inflation represent a temporary cost-push shock that left Fed credibility intact, or is there evidence that reputation began to decline? The most recent data points in Figure 3 suggest some decline in $\hat{\rho}$, is this decline statistically significant and should we worry about weakened Fed credibility going forward?

2 Minor Comments

- The authors should report confidence intervals for all smoothed and filtered reputation and cost-push shock series. Even for a fixed set of parameters, there is uncertainty arising from the filtering problem, and it would be helpful to understand how precisely these quantities are estimated.

- In Appendix C.1, the authors describe how they construct their own SPF measure using individual responses rather than the Philadelphia Fed's summary statistics, arguing that their approach is less prone to outliers. While Figure 8 shows the differences are small in most periods, reporting the robustness of the main results to using the standard FRBP measure would address potential concerns about data construction choices.

References

S. Borağan Aruoba. Term structures of inflation expectations and real interest rates. *Journal of Business & Economic Statistics*, 38(3):542–553, 2020.

Marco Del Negro, Domenico Giannone, Marc P Giannoni, and Andrea Tambalotti. Safety, liquidity, and the natural rate of interest. *Brookings Papers on Economic Activity*, 2017(1): 235–316, 2017.

Stefano Eusepi and Bruce Preston. The science of monetary policy: An imperfect knowledge perspective. *Journal of Economic Literature*, 56(1):3–59, 2018.

James D Hamilton. Why you should never use the hodrick-prescott filter. *Review of Economics and Statistics*, 100(5):831–843, 2018.

Jonathon Hazell, Juan Herreno, Emi Nakamura, and Jón Steinsson. The slope of the phillips curve: evidence from us states. *The Quarterly Journal of Economics*, 137(3):1299–1344, 2022.

Ricardo Reis. Losing the inflation anchor. *Brookings Papers on Economic Activity*, 2021(2): 307–379, 2022.